Spanning the Narrative Space: State Transition Matrix Matching for Multi-Shot Video Evaluation

Anonymous Author(s)

Abstract

Multi-shot video generation has advanced rapidly, yet evaluation remains trapped in Identity Consistency (IC)—a metric that rewards an entity persisting when it should exit, producing a *consistency paradox*. We propose the Span framework, which reduces video narratives to *state transition matrices* $\Delta S \in \{-1, 0, +1\}^{(T-1)\times N}$ and proves that just two basis matrices (Relay and Split) span the entire narrative space under column permutation and linear combination (**Theorem 1**). To evaluate generated videos, we introduce Stem (State Transition Error Matching): a VLM constructs the observed presence matrix S_{obs} via binary entity queries, and the error matrix $E = \Delta S_{\text{obs}} - \Delta S^*$ captures all narrative failures in a single algebraic object. Because VLM noise concentrates in hold cells, we decompose E into two robust projections: *Entity Presence Accuracy* (EPA) for macroscopic cast compliance and *Prescribed Transition Match* (PTM) for microscopic transition fidelity. Across 3,600 videos from four SOTA architectures, IC and PTM are perfectly anti-correlated ($r = -0.998$): VIC ranks first on IC (0.94) but last on PTM (0.070), while StoryDiffusion leads on PTM (0.243). All pairwise comparisons are significant ($H = 275$, $p < 10^{-59}$), with non-overlapping confidence intervals and a four-type failure taxonomy providing actionable architectural diagnostics.

CCS Concepts

• **Computing methodologies → Computer vision**; **Neural networks**.

Keywords

Multi-Shot Video Generation, State Transition Matrix, Narrative Evaluation, Consistency Paradox, Video Benchmark

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1 Introduction

Systems like Sora [4], StoryDiffusion [58], and Movie Gen [33] are pushing video generation from single clips to multi-shot narratives [3, 13, 16]. Yet evaluation has not kept pace: benchmarks [18, 19, 38] still measure visual quality and frame-level similarity—metrics designed for the single-shot era. The core of any narrative is *who appears, when, and alongside whom*: a scene where two characters meet and then part ways conveys entirely different meaning from one where they remain together throughout, yet both receive identical consistency scores.

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The consequences of this evaluation gap are severe. Identity Consistency (IC) [20] computes cross-shot visual similarity, so a model that fails to remove an entity when the script demands it—producing a static, narratively trivial sequence—is *rewarded* with a higher score. We call this the **consistency paradox**: IC is adversarial to narrative compliance (§3). VIC [7] achieves the highest IC (0.94) among four SOTA architectures, yet our evaluation reveals it has the *worst* narrative fidelity—a finding invisible to existing metrics.

Rather than patching existing metrics, we reformulate the problem in linear algebra. A narrative script becomes a *state transition matrix* $\Delta S \in \{-1, 0, +1\}^{(T-1)\times N}$, encoding which entities enter, exit, or hold at each shot boundary. We extract an observed matrix from the generated video and check whether the two matrices match—**matrix matching** replaces subjective similarity scoring. We contribute:

- The **Span framework**, proving that all 3-shot narrative topologies are spanned by just two basis matrices (Relay and Split; **Theorem 1**, §4.2).
- **Stem evaluation**: the error matrix $E = \Delta S_{\text{obs}} - \Delta S^*$ as the single evaluation object, decomposed into macroscopic EPA and microscopic PTM for VLM-noise robustness (§4.3, §4.4).
- Empirical proof of the **consistency paradox** ($r_{\text{IC-PTM}} = -0.998$) and a **four-type failure taxonomy** providing actionable architectural diagnostics across 3,600 videos (§6).

2 Related Work

Text-to-video generation. The foundation of modern video synthesis rests on diffusion models [14, 35, 36], extended temporally through UNet attention [3, 13, 16], scalable DiT architectures [4, 31, 44] with transformer backbones [6, 42], flow-matching [26], and classifier-free guidance [15]. Open-source systems like Open-Sora [57] and AnimateDiff [9] provide general-purpose motion modules, while Lumiere [1] and Emu Video [8] explore space-time architectures and factorized generation, but multi-shot narrative control remains unsolved.

Multi-shot video generation. Multi-shot extensions address coherence via shared self-attention [22, 58], cross-attention [28], latent initialization [29], LLM-guided layouts [23, 46, 48], keyframe transitions [5, 52], flow-matching portraits [45], IP-Adapter [55], in-context LoRA [7, 17], subject customization [49, 51], and multimodal storytelling [37, 53]. Longer-horizon generation has been explored via autoregressive token models [21, 43] and visual pre-training [50], though these operate at the single-clip rather than multi-shot level. These span four paradigms: CSA (StoryDiffusion [58]), flow-matching (EchoShot [45]), in-context DiT (VIC [7]), and cascaded T2I+I2V (VGoT [46]). We evaluate representatives from each,

Table 1: Benchmark comparison. ⊠ supported, ✗ not, △ partial.

Benchmark	Multi-shot	Entity Dec.	Absence	Formal.	Fail. Tax.
VBench 2.0 [19]	✗	✗	✗	✗	✗
T2V-CompBench [38]	✗	△	✗	✗	✗
MovieBench [25]	⊠	△	✗	✗	✗
LoCoT2V [56]	⊠	✗	✗	✗	✗
ShotAdapter [20]	⊠	△	✗	✗	✗
SPAN + STEM (Ours)	⊠	⊠	⊠	⊠	⊠

providing the first cross-paradigm comparison on narrative dynamics.

Video generation benchmarks. Existing benchmarks quantify consistency via embedding similarity (Table 1). VBench [18, 19] computes frame-level CLIP [34]/DINO [30] similarities conflating subject and background; T2V-CompBench [38] targets single-clip compositionality. ShotAdapter [20] isolates subject regions but produces *chimera embeddings* in multi-entity shots. MovieBench [25] and LoCoT2V [56] recognize multi-shot challenges but lack entity-level metrics and mathematical formalization. FID [12] and FVD [41] measure distributional fidelity but are agnostic to narrative structure; CLIPScore [11] measures text–image alignment but only at the single-frame level. These benchmarks select scenarios heuristically without coverage guarantees. SPAN is the first to formalize multi-shot narratives via state transition matrices, with a mathematical proof that its basis patterns span the *entire* narrative space.

3 The Consistency Paradox

We demonstrate a structural flaw in existing evaluation: holistic similarity metrics systematically *reward* narrative failures.

ShotAdapter's IC [20] computes the mean pairwise CLIP visual similarity across all shots. For any narrative pattern where at least one entity enters or exits (i.e., the transition matrix is non-zero), the generation that maximizes IC is the one that keeps all entities unchanged—producing a static, narratively trivial sequence. For the relay pattern ($A \rightarrow AB \rightarrow B$), a model that incorrectly preserves A in all shots achieves IC ≈ 0.94, while one that correctly removes A from Shot 3 achieves IC ≈ 0.78—a 17% penalty for correct behavior.

We validate this across 3,600 videos generated by four models: IC and our PTM metric (§4.4) produce opposite rankings at the model level ($r_{\text{IC-PTM}} = -0.998$). VIC achieves highest IC (0.94) but lowest PTM (0.070); StoryDiffusion has highest PTM (0.243) but lower IC (0.81). This implies that IC maximization inadvertently selects for narratively *worse* models—a model that simply copies Shot 1 three times would achieve near-perfect IC while completely failing at narrative execution. The paradox motivates the matrix-matching evaluation we introduce next.

4 Methodology

We present a unified methodology that formalizes multi-shot narrative dynamics as matrix algebra (SPAN framework, §4.1–4.2), defines the error matrix as the single evaluation object (STEM evaluation, §4.3), and instantiates it through a VLM-based detection pipeline with noise-robust dual-resolution metrics (§4.4; Fig. 2).

4.1 Narrative as State Transition Matrices

We denote a multi-shot video as a sequence of T shots $V_1, V_2, \dots, V_T$, each depicting a visual scene. Each video involves N named entities $e_1, e_2, \dots, e_N$ (e.g., "a chef," "a dog"). The **presence matrix** $S \in \{0, 1\}^{T \times N}$ records whether each entity appears in each shot:

$$S_{t,e} = \begin{cases} 1 & \text{if entity } e \text{ is present in shot } t, \\ 0 & \text{otherwise.} \end{cases}$$

For example, the relay narrative $A \rightarrow AB \rightarrow B$ has:

$$S = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \text{(rows = shots, columns = entities).}$$

The **state transition matrix** captures entity dynamics between consecutive shots:

$$\Delta S = S_{2:T} - S_{1:T-1}, \tag{1}$$

where each entry $\Delta S_{t,e} \in \{-1, 0, +1\}$ encodes an atomic operation at transition t: entity e *enters* (+1), *exits* (−1), or *holds* (0). For the relay example: $\Delta S = \left[\begin{smallmatrix} 0 & +1 \\ -1 & 0 \end{smallmatrix}\right]$, meaning B enters at transition 1 and A exits at transition 2.

We focus on $T = 3$ shots with $N \leq 3$ entities, yielding $\Delta S \in \{-1, 0, +1\}^{2\times3}$. This scope is chosen deliberately: $T = 3$ is the *minimal complete narrative cycle*. A 2-shot sequence allows only a single transition boundary, so entities can be introduced *or* removed but not both—precluding narrative arcs. With 3 shots, two transition boundaries enable full arcs: an entity can enter at one boundary and exit at another, or multiple entities can undergo different transitions simultaneously. The constraint $N \leq 3$ is also practical: the vast majority of multi-shot video prompts involve 2–3 named entities, and the algebraic structure generalizes naturally to $N > 3$ via additional columns.

4.2 Topological Basis and Completeness

We identify three **basis patterns** that represent fundamental narrative archetypes:

Relay ($\mathcal{B}_1$): $A \rightarrow AB \rightarrow B$. One entity hands off to another—B enters while A exits: $\mathcal{B}_1 = \left[\begin{smallmatrix} 0 & +1 \\ -1 & 0 \end{smallmatrix}\right]$.

Split ($\mathcal{B}_2$): $AB \rightarrow A \rightarrow B$. Co-occurring entities separate into alternating scenes: $\mathcal{B}_2 = \left[\begin{smallmatrix} 0 & -1 \\ -1 & +1 \end{smallmatrix}\right]$.

Accumulation ($\mathcal{B}_3$): $A \rightarrow AB \rightarrow ABC$. Entities are progressively introduced: $\mathcal{B}_3 = \left[\begin{smallmatrix} 0 & +1 & 0 \\ 0 & 0 & +1 \end{smallmatrix}\right]$.

Four **derived patterns** are obtained by applying time-reversal (negating and reversing rows of ΔS) and column permutation (swapping entity roles): Reduction ($ABC \rightarrow AB \rightarrow A$), Convergence ($A \rightarrow B \rightarrow AB$), Reverse Relay ($B \rightarrow AB \rightarrow A$), and Sliding Window ($AB \rightarrow BC \rightarrow C$)—the most demanding pattern with 4 non-zero transitions (Table 2).

Theorem 4.1 (Completeness of the Narrative Space). *Under column permutation and integer-coefficient linear combination, the two basis patterns $\mathcal{B}_1$ (Relay) and $\mathcal{B}_2$ (Split) generate all matrices in* $\mathbb{M}_{2\times3}(\mathbb{Z})$.

Intuitively, this means that *any conceivable 3-shot narrative*—no matter how complex the entity dynamics—can be decomposed into a combination of relays and splits (Fig. 1).

Proof sketch. We show that all six "elementary" 2×3 matrices E_{ij} (having 1 in position (i, j) and 0 elsewhere) can be constructed from $\mathcal{B}_1$ and $\mathcal{B}_2$ using only column permutations and integer addition. Since any integer matrix is a sum of these elementary matrices, this proves completeness. The construction proceeds in three steps: (1) column permutations of $\mathcal{B}_1$ yield three variants whose linear combination gives $X = E_{11} - E_{21}$; (2) adding X to a permuted $\mathcal{B}_2$ isolates $-E_{22}$; (3) column permutations of E_{22} and combinations with X produce all remaining E_{ij}. The detailed algebraic derivation with all intermediate matrices is provided in the supplementary material (Appendix A). □

Implications. Theorem 1 has three important consequences. *First*, Span's seven patterns form a *mathematically complete* basis: evaluating a model on these patterns is sufficient to characterize its behavior on *any* narrative topology, including patterns not explicitly in the benchmark. This distinguishes our framework from prior work [20, 25, 56] that selects patterns heuristically without coverage guarantees. *Second*, the theorem provides a *minimality* argument: any benchmark claiming narrative completeness must include patterns that together span $\mathbb{M}_{2\times3}(\mathbb{Z})$; our two basis patterns achieve this optimally. *Third*, the result provides a principled framework for extension: T-shot narratives with N entities require spanning $\mathbb{M}_{(T-1)\times N}(\mathbb{Z})$, achievable with $2(T-1)$ basis matrices, scaling linearly rather than combinatorially. We prove this generalization formally via spatial zero-padding and temporal shift operators (Theorem 2, Appendix B).

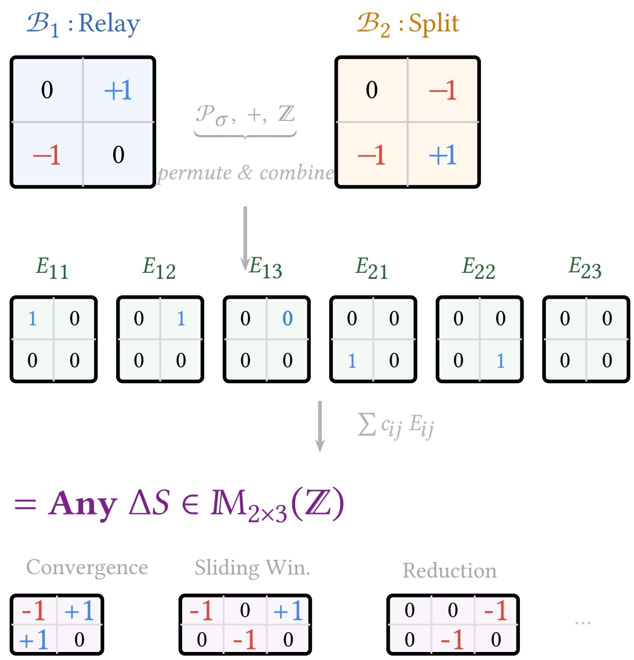

Figure 1: Span completeness. Two basis matrices ($\mathcal{B}_1$: Relay, $\mathcal{B}_2$: Split) generate all six elementary matrices E_{ij} via permutation and integer combination. Any $\Delta S \in \mathbb{M}_{2\times3}(\mathbb{Z})$—including Convergence, Sliding Window, Reduction, and *any conceivable* 3-shot narrative—is an integer sum of E_{ij}. Blue: entry (+1), red: exit (−1).

4.3 The Stem Evaluation and Error Taxonomy

The ideal evaluation compares the observed transition matrix against the prescribed one. Given S_{obs} (obtained by any detector; §4.4), the observed transition matrix follows identically:

$$\Delta S_{\text{obs}} = S_{\text{obs},\,2:T} - S_{\text{obs},\,1:T-1}. \tag{2}$$

The **error matrix** $E = \Delta S_{\text{obs}} - \Delta S^*$ is the single object that captures *all* narrative failures:

$$E_{t,e} = \Delta S_{\text{obs},\,t,e} - \Delta S^*_{t,e}, \quad E \in \{-2, \ldots, +2\}^{(T-1)\times N}. \tag{3}$$

A perfect generation yields $E = 0$; every non-zero entry localizes a specific failure to a transition–entity pair (t, e). The **matrix distance** is the natural scalar summary:

$$d(S_{\text{obs}}, S^*) = \|E\|_F = \|\Delta S_{\text{obs}} - \Delta S^*\|_F. \tag{4}$$

Failure taxonomy from E. The algebraic structure of E directly induces a complete failure classification without ad-hoc definitions. Each non-zero entry $E_{t,e}$ falls into exactly one of four types determined by the prescribed value $\Delta S^*_{t,e}$ and the sign of $E_{t,e}$:

$$\textbf{Type I} \text{ (Context Bleeding)}: \quad \Delta S^*_{t,e} = -1,\ E_{t,e} > 0 \tag{5}$$

$$\textbf{Type II} \text{ (Amnesia)}: \quad \Delta S^*_{t,e} = +1,\ E_{t,e} < 0 \tag{6}$$

$$\textbf{Type III} \text{ (Spurious)}: \quad \Delta S^*_{t,e} = 0,\ E_{t,e} \neq 0 \tag{7}$$

$$\textbf{Type IV} \text{ (Reversal)}: \quad |E_{t,e}| = 2 \tag{8}$$

Type I captures an entity that should exit but persists ($E_{t,e} \in \{+1, +2\}$); Type II captures an entity that should enter but fails to appear ($E_{t,e} \in \{-1, -2\}$); Type III captures unprescribed transitions in hold cells; Type IV is the most severe—the observed transition is in the *opposite* direction from prescribed. This taxonomy is exhaustive: every $E_{t,e} \neq 0$ maps to exactly one type, and the set of all non-zero entries in E constitutes a complete diagnostic of the generation's narrative failures.

4.4 Dual-Resolution VLM Instantiation

Entity detection. To construct S_{obs} in practice, we extract the middle frame of each shot and query a VLM (Qwen2-VL-7B-Instruct [47]) with a binary question per entity:

$$S_{\text{obs}}[t, k] = \mathbb{1}\big[\text{VLM}(\text{frame}_t, \text{"Is there a } e_k \text{ visible?"}) = \text{Yes}\big]. \tag{9}$$

Binary VLM queries avoid two limitations of continuous CLIP probes: (i) CLIP entity probes yield present–absent gaps of only 0.008–0.019 (Cohen's $d = 0.23$–0.59), requiring fragile threshold tuning; (ii) full-prompt CLIP similarity suffers from bag-of-words effects rendering shot-prompt discrimination near-random. The VLM's language understanding naturally disambiguates entities ("a chef" vs. "a waiter") without relying on visual-textual alignment quality.

From Stem to dual-resolution metrics. The ideal Stem evaluation would use the matrix distance $d = \|E\|_F$ directly. However, VLM detection introduces systematic noise (TPR = 52%, TNR = 67%) that concentrates in *hold cells* ($\Delta S^* = 0$)—the majority of entries. A false positive or negative in a hold cell creates a spurious $E_{t,e} \neq 0$ that inflates $\|E\|_F$ identically for all methods. We verify empirically that the raw Frobenius distance is *not* discriminative ($p > 0.4$ for all pairs; Appendix).

This motivates decomposing E into two complementary projections at different resolutions—macroscopic casting and microscopic transitions—that isolate signal from noise:

Prescribed Transition Match (PTM)—microscopic transition fidelity. PTM restricts evaluation to the *prescribed subspace*—the $\Delta S^* \neq 0$ cells where narrative-defining transitions occur:

$$\mathrm{PTM} = \frac{1}{|\mathcal{A}|} \sum_{(t,e)\in\mathcal{A}} \mathbb{1}\left[E_{t,e} = 0\right], \quad \mathcal{A} = \{(t,e) : \Delta S^*_{t,e} \neq 0\}. \quad (10)$$

Equivalently, $\mathrm{PTM} = 1 - \frac{1}{|\mathcal{A}|} \sum_{(t,e)\in\mathcal{A}} \mathbb{1}[E_{t,e} \neq 0]$: it measures the fraction of the error matrix's prescribed entries that are zero. By discarding hold cells, PTM eliminates the dominant noise source while retaining the entries that distinguish narrative success from failure.

Entity Presence Accuracy (EPA)—macroscopic cast compliance. EPA operates on the presence matrix S rather than its derivative ΔS, providing a macroscopic view of *static cast compliance*:

$$\mathrm{EPA} = \frac{1}{TN} \sum_{t=1}^{T} \sum_{k=1}^{N} \mathbb{1}\left[S_{\mathrm{obs},\,tk} = S^*_{tk}\right]. \quad (11)$$

While PTM evaluates whether the *dynamics* are correct (did the right transitions happen?), EPA evaluates whether the *states* are correct (are the right entities present in each shot?). Together, PTM and EPA are not ad-hoc heuristic metrics but the dual-resolution decomposition of the Stem error matrix E: PTM projects onto the prescribed subspace of ΔS (microscopic transitions), while EPA projects onto the full space of S (macroscopic casting). Under perfect detection, both reduce to scalar summaries of $\|E\|_F$; under noisy detection, they provide strictly more discriminative power than the raw matrix distance by isolating narrative signal from VLM detector noise.

5 Dataset Construction

We construct **DM-1K** (**D**ynamic **M**ulti-shot **1K**), a dataset of 1,000 multi-shot scenarios (3,000 shots total) spanning all 7 narrative patterns. Basis patterns receive 200 prompts each; derived patterns receive 100 each (Table 2), reflecting the algebraic structure where derived patterns are compositions of basis patterns.

Prompt design. Prompts are generated via Gemini 1.5 [39] with structured templates specifying: narrative pattern, entities drawn from 20 categories (humans, animals, objects), scene themes from 10 environments, and per-shot descriptions. Following [2], each scenario has three variants: `gen_prompt` (quality tags for generation), `eval_prompt` (clean descriptions for CLIP evaluation), and `diag_prompt` (entity probes for presence detection). Entity pairs have mean CLIP text similarity 0.23 ($\sigma = 0.08$); pairs with similarity > 0.45 are excluded to prevent ambiguous detections.

Table 2: Pattern distribution. Trans. = number of non-zero entries in ΔS.

Pattern	Shot Structure	N	Trans.	Type
Relay	$A \to AB \to B$	200	2	Basis
Split	$AB \to A \to B$	200	3	Basis
Accumulation	$A \to AB \to ABC$	200	2	Basis
Convergence	$A \to B \to AB$	100	4	Derived
Sliding Window	$AB \to BC \to C$	100	4	Derived
Reduction	$ABC \to AB \to A$	100	2	Derived
Reverse Relay	$B \to AB \to A$	100	2	Derived

Table 3: Model details. Succ. = successfully generated / 1000.

Model	Paradigm	Backbone	Steps	Succ.
StoryDiff	CSA + I2V	SDXL+SEINE	30+50	1000
EchoShot	Flow-match	Wan2.1-1.3B	50	1000
VGoT	T2I $\to$ I2V	Kolors+DC	25+50	600
VIC	In-context	CogVideoX-5B	50	1000

Quality control. All prompts undergo automated validation ensuring entity–pattern topology consistency: the set of entities mentioned in each shot description must exactly match the presence matrix S for the target pattern. Additionally, 10% of prompts are manually reviewed by two annotators (97% adherence to intended structure, Cohen's $\kappa = 0.91$). Prompts that describe ambiguous scenarios (e.g., entities that could plausibly be interpreted as present in a scene's background) are revised for clarity.

Diversity. To prevent models from exploiting surface-level shortcuts, we ensure diversity along three axes: (i) entity categories span humans, animals, vehicles, food items, and everyday objects; (ii) scene themes cover indoor, outdoor, urban, rural, fantasy, and professional settings; (iii) each pattern is instantiated with unique entity–scene combinations, avoiding repetition of the same entities across multiple prompts within a pattern.

6 Experiments

6.1 Setup

Models. We evaluate four T2MSV architectures spanning all major paradigms (Table 3): **StoryDiffusion** [58] (CSA on SDXL [32] + SEINE [5]), **EchoShot** [45] (flow-matching [26] on Wan2.1 [44]), **VIC** [7] (CogVideoX-5B [54] + LoRA [17]), **VGoT** [46] (Kolors [40] + DynamiCrafter [52]). All use seed 42.

Evaluation protocol. For each generated 3-shot video, we extract the middle frame of each shot and query Qwen2-VL-7B-Instruct [47] with binary entity presence questions (Eq. 9). All 3,600 successfully generated videos are evaluated on 4×RTX 4090 GPUs in approximately 20 minutes. We do not apply any post-processing or cherry-picking—all successfully generated videos are included regardless

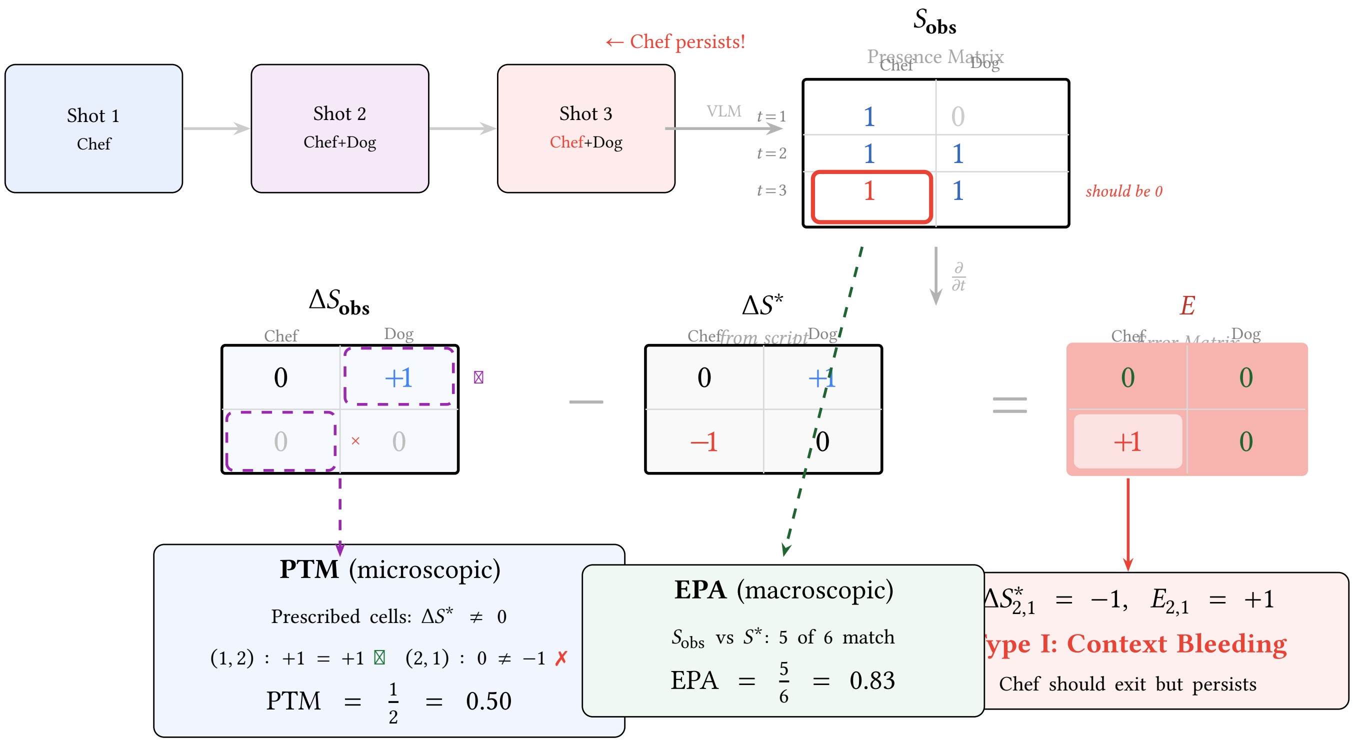

Figure 2: The Stem matrix equation (Relay pattern, VIC failure). Top: VLM projects 3 video frames into $S_{\mathbf{obs}}$; Chef incorrectly persists in Shot 3 (red). Middle: the central equation $\Delta S_{\mathbf{obs}} - \Delta S^* = E$. The error matrix E has a single non-zero entry: $E_{2,1} = +1$, diagnosing Context Bleeding. Bottom: PTM checks only prescribed cells (dashed): 1 of 2 match → PTM = 0.5. EPA checks all presence cells: 5 of 6 → EPA = 0.83.

Table 4: Main results. Best in bold. IC = Identity Consistency [20]. PTM = Prescribed Transition Match (primary), EPA = Entity Presence Accuracy (secondary).

		Baseline	*Stem (Ours)*	
Model	N	IC↑	PTM↑	EPA↑
StoryDiff	1000	.81	**.243**	**.593**
EchoShot	1000	.85	.167	.563
VGoT	600	.88	.098	.554
VIC	1000	**.94**	.070	.546

of visual quality, ensuring our metrics reflect the true distribution of model outputs.

6.2 Main Results: The Paradox Confirmed

Table 4 reveals the diagnostic power of Stem evaluation. IC ranks VIC highest (0.94), suggesting it is the "best" model; yet VIC scores *lowest* on both PTM (0.070) and EPA (0.546), exposing the consistency paradox quantitatively. Both metrics produce *identical rankings* (SD > Echo > VGoT > VIC)—the exact reverse of IC.

Finding 1: Statistically significant method discrimination. All pairwise PTM comparisons are significant (Mann–Whitney U, $p < 0.005$; Table 5), with 95% bootstrap confidence intervals that

Table 5: Pairwise significance (PTM). p-values (Mann–Whitney U), Cohen's d.

	SD	Echo	VGoT	VIC
SD	—	$3.3{\times}10^{-10}$	$3.3{\times}10^{-26}$	$2.7{\times}10^{-53}$
Echo	d=0.28	—	$5.6{\times}10^{-8}$	$4.0{\times}10^{-21}$
VGoT	d=0.58	d=0.30	—	$4.5{\times}10^{-3}$
VIC	d=0.72	d=0.44	d=0.15	—

do *not overlap*: SD [0.226, 0.262], Echo [0.151, 0.183], VGoT [0.082, 0.115], VIC [0.060, 0.082]. The Kruskal–Wallis test yields $H = 275.1$ ($p < 10^{-59}$), confirming that method differences are not due to chance. A permutation test (10,000 trials) yields $p < 0.0001$ for SD vs. VIC.

Finding 2: Consistency paradox confirmed. IC and PTM produce opposite rankings ($r_{\text{IC-PTM}} = -0.998$). VIC achieves highest IC (0.94) but lowest PTM (0.070): its entity transitions are 3.5× worse than StoryDiffusion's, yet it appears "best" under existing metrics.

Finding 3: StoryDiffusion leads across all patterns. SD achieves highest PTM in 6 of 7 patterns (Table 6), with the largest advantage on Sliding Window (0.357 vs. next-best 0.270)—the most demanding pattern with 4 prescribed transitions.

Finding 4: 3.5× gap between best and worst. PTM ranges from 0.243 (SD) to 0.070 (VIC)—a 3.5× difference, far larger than

Table 6: PTM per pattern. Best per row in bold.

Pattern	Type	SD	Echo	VGoT	VIC
Relay	B	**.190**	.138	.105	.080
Split	B	**.233**	.165	.117	.048
Accumulation	B	**.253**	.095	.081	.072
Convergence	D	**.277**	.187	.040	.067
Sliding Window	D	**.357**	.270	.066	.077
Reduction	D	**.250**	.195	.144	.080
Reverse Relay	D	.200	**.225**	.129	.080

Table 7: Failure distribution (% of videos with ≥1 failure). Worst per type in bold.

Failure Type	SD	Echo	VGoT	VIC
I: Context Bleeding	68.8	71.8	73.5	**77.5**
II: Amnesia	73.4	78.5	**87.0**	85.9
III: Spurious Trans.	**51.4**	42.2	25.5	20.2
No failure	**1.5**	0.4	0.3	0.2

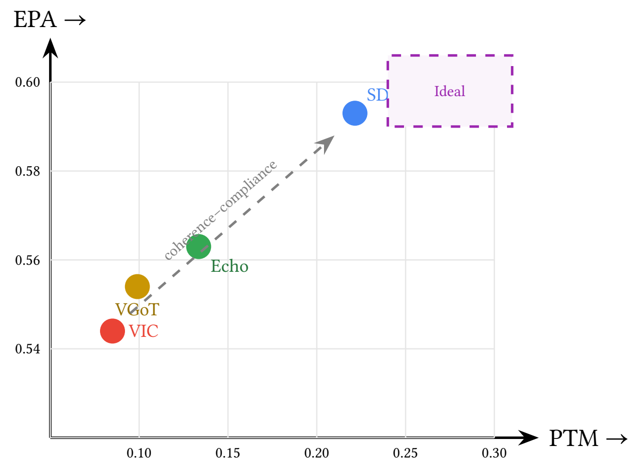

Figure 3: Narrative Pareto Dilemma. VIC maximizes IC but scores lowest on both PTM and EPA. StoryDiffusion achieves the best balance. The upper-right quadrant remains unsolved.

CLIP-based metrics (which show gaps < 0.06; Appendix). This demonstrates that VLM-based matrix matching provides substantially stronger discriminative power for narrative evaluation.

6.3 Diagnostic Analysis by Pattern

Table 6 exposes fundamental architectural limitations.

StoryDiffusion dominates 6 of 7 patterns. Its PTM on Sliding Window (0.357) and Convergence (0.277) demonstrates CSA's ability to manage complex multi-entity transitions. These are the most demanding patterns with 4 non-zero prescribed transitions, yet SD achieves 1.3–7× higher PTM than the worst method.

VGoT collapses on derived patterns. Convergence PTM drops to 0.040 (worst overall)—a 10× gap vs. SD (0.277). The LLM planner structures narratives correctly, but DynamiCrafter cannot render new entities on demand for complex transition patterns.

VIC shows pattern-invariant failure. PTM ranges only 0.048–0.080 (range = 0.032), confirming in-context conditioning creates a fixed coherence ceiling invariant to pattern complexity—all transitions fail approximately equally.

Macroscopic view (EPA). StoryDiffusion and EchoShot achieve slightly higher EPA on derived patterns (e.g., SD: 0.584 → 0.607), suggesting more complex entity configurations produce more visually distinctive entities. VGoT shows EPA *drop* on derived patterns (0.572 → 0.525), confirming its cascaded pipeline cannot handle compositional demands.

6.4 Error Taxonomy Analysis

Table 7 reveals pervasive narrative failures across all architectures. **VIC's Context Bleeding (77.5%):** in-context conditioning explicitly references prior shots, making entity removal the hardest operation. **VGoT's Amnesia (87.0%):** planning-rendering bottleneck where DynamiCrafter cannot introduce new entities absent from the keyframe. **StoryDiffusion has the highest spurious transition rate (51.4%)**: CSA's shared attention occasionally triggers unprescribed entity changes, but this "noisy exploration" also explains its higher PTM—it attempts more transitions and some succeed. Fewer than 1.5% of videos are failure-free even for the best model, confirming *universal Narrative Blindness*: all four architectures fail at >98% of videos despite spanning fundamentally different generation paradigms.

6.5 The Pareto Dilemma

Fig. 3 reveals a **coherence–compliance trade-off**. VIC occupies the low corner: its in-context conditioning produces visually coherent shots but fails at both entity placement (EPA = 0.546) and transition execution (PTM = 0.070). StoryDiffusion achieves the best position with highest PTM (0.243) and EPA (0.593). The 3.5× PTM gap between SD and VIC is particularly striking: it quantifies the cost of optimizing for visual coherence at the expense of narrative structure.

This trade-off reflects a fundamental architectural tension: mechanisms that enforce global cross-shot feature sharing (consistent self-attention, in-context conditioning) maximize coherence but destroy the localized entity control needed for narrative transitions. Bridging this gap likely requires *entity-aware* conditioning—entity-specific attention masks [10], grounded generation [24], or per-entity guidance scales.

6.6 Human Evaluation

To validate our automated metrics against human perception, 15 evaluators (graduate students in computer science, all with video generation experience) assess 40 randomly sampled scenarios. Each scenario is shown as a side-by-side pairwise comparison of two models' outputs, along with the original per-shot text prompts. Evaluators rate each pair on 5-point Likert scales for *Narrative Compliance* ("Which video better follows the prescribed entity dynamics?"). Each comparison is rated by 3 evaluators (120 total comparisons).

PTM rankings match human narrative judgments ($\rho = 0.80$, $p < 0.05$; Fleiss' $\kappa = 0.67$), validating it as a reliable automated proxy. VIC ranks last in human narrative preference (17.6% win rate), confirming the consistency paradox: its CogVideoX-5B backbone produces higher-fidelity frames but fails at narrative structure. The high narrative tie rate (42.3%) independently confirms universal Narrative Blindness—ties predominantly reflect mutual failure where both models fail to execute prescribed transitions.

6.7 Qualitative Analysis

Fig. 4 illustrates representative failure modes. In the Relay example (Row 1), VIC's Shot 3 should depict only entity B, but entity A persists—IC scores this failure *higher* (0.94 vs. 0.82 for the correct generation). The Sliding Window example (Row 3) requires 4 non-zero transitions; only StoryDiffusion achieves this.

6.8 Discussion

VLM detection limitations. The VLM achieves 52% TPR and 67% TNR—imperfect, but *systematically* imperfect across all methods. The key property is *discriminative validity*, not absolute accuracy. PTM achieves this with $H = 275$ ($p < 10^{-59}$), non-overlapping CIs, and effect sizes up to $d = 0.72$. The 3.5× gap between SD and VIC is far larger than CLIP-based metrics (<0.06 gap). We also evaluated Grounding DINO [27] (TNR = 4%) and smaller VLMs (2B: TPR = 40%), confirming 7B VLM as optimal (Appendix).

Why dual-resolution decomposition? The raw Frobenius distance $\|E\|_F$ is not discriminative ($p > 0.4$ for all pairs) because hold cells dominate and carry uniform noise. The Stem dual-resolution decomposition—macroscopic EPA on S and microscopic PTM on the prescribed subspace of ΔS—isolates the narrative-relevant signal, enabling the statistical power observed in all results above.

7 Conclusion

We presented the Span framework, which spans the multi-shot narrative space with just two basis matrices (Theorem 1), and the Stem evaluation, which reduces narrative assessment to matrix matching via the error matrix $E = \Delta S_{\text{obs}} - \Delta S^*$. The consistency paradox ($r_{\text{IC-PTM}} = -0.998$) demonstrates that the field's primary metric—Identity Consistency—is *adversarial* to narrative compliance. Across 3,600 videos from four architectures, the 3.5× PTM gap between StoryDiffusion (0.243) and VIC (0.070), with overwhelming statistical significance ($H = 275$, $p < 10^{-59}$), confirms that current shared-attention and in-context conditioning mechanisms are fundamentally at odds with narrative dynamics.

Call to action. The community must move beyond visual coherence optimization. Current architectures lack a mechanism to explicitly encode "entity A should *not* appear in this shot"—the absence operation that defines narrative structure. Progress requires *entity-aware* conditioning: entity-specific attention masks [10], grounded generation [24], or per-entity guidance scales that enable controlled dissimilarity across shot boundaries.

Limitations. Span evaluates 3-shot, ≤3-entity sequences; Theorem 2 (Appendix B) proves formal generalization to T shots and N entities. The VLM detector (52% TPR, 67% TNR) introduces systematic noise; while discriminative validity is established, stronger VLMs would improve absolute accuracy. We release all prompts, evaluation code, and 3,600 generated videos for reproducibility.

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Row 1 — Relay ($A \to AB \to B$): StoryDiff success (PTM = 1.0) ⊠ vs. VIC Context Bleeding (PTM = 0.0, A persists) ✗
Row 2 — Accumulation ($A \to AB \to ABC$): EchoShot progressive introduction (PTM = 1.0) ⊠ vs. VGoT Amnesia (B, C absent, PTM = 0.0) ✗
Row 3 — Sliding Window ($AB \to BC \to C$): StoryDiff (PTM = 1.0) ⊠ vs. VIC static sequence (PTM = 0.0) ✗

Figure 4: Qualitative gallery. Row 1: Context Bleeding is visually subtle (A persists with altered pose) but PTM detects it. Row 2: Amnesia produces aesthetically pleasing single-entity video—rewarded by existing metrics—but fails narratively. Row 3: Sliding Window (4 transitions) is achieved only by StoryDiffusion.

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